HOTS

Which is more dangerous – bacteria or Virus?

Prion

Prions are small proteinaceous infectious agents without genetic material. These are responsible for a number of degenerative brain diseases (Example: Creutzfeldt) and hereditary dementia.

10.2 Cultivation of Viruses

Viruses are **obligate intracellular** parasites; they cannot be grown on any inanimate culture medium. **Three methods** are employed for the cultivation of viruses – **inoculation into animals, embryonated eggs and tissue culture or cell culture**.

i. Animal Inoculation

The earliest method for the cultivation of viruses causing human diseases was inoculation into human volunteers. Monkeys were used for the isolation of the polio virus by **Landsteiner and Popper (1909)**. The embryonated hen's egg was first used for cultivation of viruses by **Good pasture (1931)**. The embryonated egg offers several sites for the cultivation of viruses. Non human primates provide the only method for virus cultivation. Mice are most widely employed animals in Virology.

ii. Embryonated Eggs

a. Chorioallantonic Membrane (CAM)

Inoculation on **the chorioallantonic membrane** produces visible lesions (pocks). Different viruses have different pock morphology. Example: varcola or vaccinia

b. Allantonic Cavity

Inoculation on the **allantonic cavity** provides a rich yield of influenza and some paramyxo viruses.

c. Amniotic Sac

Inoculation into the **amniotic sac** is for the primary isolation of the influenza virus.

d. Yolk Sac

Inoculation into the **yolk sac** is for the cultivation of some viruses like Chlamydiae and Rickettsiae.

Allantonic inoculation is employed for growing influenza virus for vaccine production (Figure 10.3).

iii. Tissue Culture

First tissue culture in Virology was maintained by **Steinhardt and colleagues (1913)** for the vaccinia virus in fragments of rabbit cornea. Bacterial contamination was the major limitation. Different types of culture used are:

a. Organ culture

Small bits of organs can be maintained, used for the isolation of some viruses. Example: Corona virus (respiratory pathogen) cultured on tracheal ring organ culture.

b. Explant culture

Fragments of minced tissue are grown as 'explants'. This is also known as tissue culture. Example: Adeno virus cultured on Adenoid tissue explants.

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